WORKED SOLUTION

STEP II 2012, Question 2

Pure • Composition of Polynomials

STEP II 2012, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q2

The key idea

Degrees multiply under composition, so a few iterations of p quickly produce a degree far too large to match the right-hand side. Comparing degrees pins down the degree of p; after that, writing p out with unknown coefficients turns each part into a short piece of algebra. In (ii) there is a neat shortcut: $\mathrm{p}(\mathrm{p}(x))$ is itself a quadratic expression *in* $\mathrm{p}(x)$, so the whole left-hand side is a quadratic in $\mathrm{p}(x)$, and completing the square finishes the job.

Solution

The degree of a composition

Write $\mathrm{p}(x) = a_m x^m + \dots$ and $\mathrm{q}(x) = b_n x^n + \dots$ with $a_m, b_n \neq 0$. In $\mathrm{p}(\mathrm{q}(x)) = a_m[\mathrm{q}(x)]^m + a_{m-1}[\mathrm{q}(x)]^{m-1} + \dots$, the term $a_m[\mathrm{q}(x)]^m$ has degree mn and leading coefficient $a_m b_n^m \neq 0$, while every other term has degree at most $(m-1)n < mn$ (when $n \geq 1$). So nothing cancels the top term, and

$$\deg \mathrm{p}(\mathrm{q}(x)) = mn.$$

(If $n = 0$ then $\mathrm{p}(\mathrm{q}(x))$ is a constant, and $mn = 0$ again.)

Part (i)

Let p have degree m.

If $m \geq 2$, then by the result above $\mathrm{p}(\mathrm{p}(\mathrm{p}(x)))$ has degree m^3, while $3\mathrm{p}(x)$ has degree $m < m^3$. So the left-hand side has degree $m^3 \geq 8$, but the right-hand side $-2x$ has degree 1. Impossible.

If $m = 0$, then $\mathrm{p}(x) = c$ for a constant c, and the left-hand side is the constant $c - 3c = -2c$, which cannot equal $-2x$ for all x. Impossible.

So $m = 1$. Write $\mathrm{p}(x) = ax + b$ with $a \neq 0$. Then

$$\mathrm{p}(\mathrm{p}(x)) = a(ax + b) + b = a^2x + (a + 1)b,$$
$$\mathrm{p}(\mathrm{p}(\mathrm{p}(x))) = a^3x + (a^2 + a + 1)b,$$

and the equation becomes

$$(a^3 - 3a)x + (a^2 + a + 1 - 3)b = -2x \quad \text{for all } x.$$

Two polynomials agree for all x only if their coefficients agree, so

$$a^3 - 3a + 2 = 0 \qquad \text{and} \qquad (a^2 + a - 2)\, b = 0.$$

Since $a = 1$ is a root of the first, it factorises as $(a - 1)^2(a + 2) = 0$, giving $a = 1$ or $a = -2$. The second condition is $(a - 1)(a + 2)b = 0$, which holds automatically for both of these values, whatever b is. Hence the solutions are exactly

$$\mathrm{p}(x) = x + b \qquad \text{and} \qquad \mathrm{p}(x) = -2x + b, \qquad b \text{ any constant.}$$

As a check on the second family: $\mathrm{p}(\mathrm{p}(x)) = 4x - b$ and $\mathrm{p}(\mathrm{p}(\mathrm{p}(x))) = -8x + 3b$, so the left-hand side is $-8x + 3b + 6x - 3b = -2x$.

Part (ii)

Let p have degree m. The three terms on the left have degrees m^2, $2m$ and m.

- If $m \geq 3$, then $m^2 > 2m > m$, so the left-hand side has degree $m^2 \geq 9 \neq 4$.
- If $m = 1$, the left-hand side has degree at most 2; if $m = 0$ it is constant. Neither can equal x^4.

So $m = 2$. (Here $m^2 = 2m = 4$, so the top terms of $2\mathrm{p}(\mathrm{p}(x))$ and $3[\mathrm{p}(x)]^2$ have the same degree, and we must check their coefficients.)

Write $\mathrm{p}(x) = ax^2 + bx + c$ with $a \neq 0$, and to save writing let P stand for $\mathrm{p}(x)$.

> **Ask yourself** Do we really have to expand $\mathrm{p}(\mathrm{p}(x))$ in full?

No. By definition $\mathrm{p}(\mathrm{p}(x)) = aP^2 + bP + c$, so the left-hand side is

$$2(aP^2 + bP + c) + 3P^2 - 4P = (2a + 3)P^2 + (2b - 4)P + 2c,$$

a quadratic in P. Its x^4 coefficient is $(2a + 3)a^2$, since P^2 has leading term a^2x^4 and P has degree only 2. Matching with x^4 gives

$$2a^3 + 3a^2 - 1 = 0, \qquad \text{that is,} \qquad (a + 1)^2(2a - 1) = 0,$$

(spot the root $a = -1$ and divide), so $a = -1$ or $a = \frac{1}{2}$.

We now use a simple fact: if $S(x)$ is a polynomial and $S(x)^2 = x^4 + K$ for a constant K, then $(S - x^2)(S + x^2) = K$. If neither factor were the zero polynomial, at least one of

them would have degree 2 when S has leading term $\pm x^2$, and the product could not be constant. So, for S with leading term $\pm x^2$, one factor is zero and $S = \pm x^2$ exactly.

Case $a = -1$. Then $2a + 3 = 1$, and completing the square,

$$P^2 + (2b-4)P + 2c = \big(P + b - 2\big)^2 - (b-2)^2 + 2c = x^4.$$

So $S = P + b - 2 = -x^2 + bx + (c + b - 2)$ satisfies $S^2 = x^4 + K$ with K constant, and has leading term $-x^2$. Then $S + x^2 = bx + (c + b - 2)$ has degree less than 2 while $S - x^2$ has degree 2, so the fact forces $S + x^2 = 0$: that is, $b = 0$ and $c + b - 2 = 0$. So $b = 0$, $c = 2$, and then $K = (b-2)^2 - 2c = 0$, consistent. This gives

$$\mathrm{p}(x) = 2 - x^2.$$

Case $a = \frac{1}{2}$. Then $2a + 3 = 4$, and

$$4P^2 + (2b-4)P + 2c = \left(2P + \tfrac{b-2}{2}\right)^2 - \tfrac{(b-2)^2}{4} + 2c = x^4.$$

Now $S = 2P + \frac{b-2}{2} = x^2 + 2bx + 2c + \frac{b-2}{2}$ has leading term x^2, so in the same way $S - x^2 = 0$: $b = 0$ and $2c - 1 = 0$. So $b = 0$, $c = \frac{1}{2}$, and $K = \frac{(b-2)^2}{4} - 2c = 1 - 1 = 0$, consistent. This gives

$$\mathrm{p}(x) = \tfrac{1}{2}(x^2 + 1).$$

Both really are solutions, since every step can be reversed, but it is reassuring to check directly. For $\mathrm{p}(x) = 2 - x^2$, $\mathrm{p}(\mathrm{p}(x)) = 2 - (2 - x^2)^2 = -x^4 + 4x^2 - 2$, and

$$2(-x^4 + 4x^2 - 2) + 3(x^4 - 4x^2 + 4) - 4(2 - x^2) = x^4.$$

For $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$, write $u = x^2 + 1$; then $2\mathrm{p}(\mathrm{p}(x)) = \frac{1}{4}u^2 + 1$, and the left-hand side is $\frac{1}{4}u^2 + 1 + \frac{3}{4}u^2 - 2u = (u-1)^2 = x^4$.

So the polynomials satisfying the equation in (ii) are exactly $\mathrm{p}(x) = 2 - x^2$ and $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$.

Checks

- In (i), try $b = 1$: $\mathrm{p}(x) = x + 1$ gives $(x + 3) - 3(x + 1) = -2x$.
- In (ii), put $x = 1$: for $\mathrm{p}(x) = 2 - x^2$ we get $\mathrm{p}(1) = 1$, $\mathrm{p}(\mathrm{p}(1)) = 1$, and $2 + 3 - 4 = 1 = 1^4$. For $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$ at $x = 3$: $\mathrm{p}(3) = 5$, $\mathrm{p}(5) = 13$, and $26 + 75 - 20 = 81 = 3^4$.
- We also checked with exact arithmetic that both families in (i) work for many random b, and that a search over a grid of quadratics with b, c in steps of $\frac{1}{4}$ finds only the two solutions in (ii).

Common slips

- In (ii), assuming the degree of the left-hand side is simply m^2: when $m = 2$ the terms $2\mathrm{p}(\mathrm{p}(x))$ and $3[\mathrm{p}(x)]^2$ both have degree 4, and the possibility that they cancel is exactly why a must be found from $2a^3 + 3a^2 = 1$.
- Forgetting the case $m = 0$ when explaining why the degree must be 1 in (i).
- In (i), stopping at $a = 1$ or $a = -2$ and fixing a particular b: the constant term condition holds for every b.

Beyond STEP

The rule $\deg(\mathrm{p} \circ \mathrm{q}) = \deg \mathrm{p} \cdot \deg \mathrm{q}$, like $\deg(\mathrm{pq}) = \deg \mathrm{p} + \deg \mathrm{q}$, relies on there being no zero divisors among the coefficients: in a ring where two non-zero numbers can multiply to zero, leading terms can vanish and both rules fail. Functional equations such as $\mathrm{p}(\mathrm{p}(x)) = \ldots$ also lead into the study of iterated maps and dynamics. For polynomial rings and when degree arguments work, see Allenby, *Rings, Fields and Groups*.

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